CONE5112/w Revision Guide

Computer Networks 1B · Kurose & Ross (2022), *Computer Networking: A Top-Down Approach*, 8th ed.

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How to use this guide

The exam is a 2-hour, closed-book, 100-mark paper covering all six learning units, mixing theory with scenario-based application. This guide condenses each unit into the concepts most likely to be examined, then tests you with questions at three levels.

Each unit follows the same layout

1. **Key concepts** – condensed notes mapped to the learning outcomes (LOs).
2. **Key terms** – definitions to learn word-perfect.
3. **Worked example** – a calculation or scenario solved step by step.
4. **Common mistakes** – where students usually lose marks.
5. **Revision questions** – Level 1 (recall), Level 2 (explain and apply), Level 3 (analyse, evaluate, scenario).

Suggested four-week revision plan

Week	Focus	Target
1	LU1 and LU2	Notes summarised; all Level 1 questions answered from memory
2	LU3 and LU4	All calculations practised twice; subnetting under time pressure
3	LU5 and LU6	Dijkstra, distance-vector and CRC practised; Level 2 questions
4	All units	Level 3 scenario questions under exam timing; formula sheet from memory

Study method that works for this module

- Close the book and write the answer first, then check. Recognition is not recall.
- For every protocol, be able to state: which layer, what problem it solves, how it works, and one limitation.

- Draw diagrams (handshakes, timelines, topologies). Labelled diagrams earn marks in long questions.
- Practise each calculation type until you can do it in under 8 minutes.
- Allocate about 1.2 minutes per mark in the exam, and one developed point per mark.

Learning Unit 1: Computer Networks, the Internet, Wireless and Mobile Networks

Kurose & Ross Chapters 1 and 7 · Academic weeks 1–2 · LO1–LO10.

Key concepts

The Internet as a network of networks (LO1)

- *Nuts-and-bolts view:* billions of hosts (end systems) connected by communication links and packet switches (routers and link-layer switches), running protocols such as TCP, IP and HTTP.
- *Services view:* an infrastructure that provides services to distributed applications through a socket interface.
- *Structure:* access ISPs connect to regional ISPs, which connect to tier-1 ISPs. Internet Exchange Points (IXPs) let ISPs peer directly. Content-provider networks (e.g. Google) run their own global networks to bypass upper tiers.
- A **protocol** defines the format and order of messages exchanged and the actions taken on sending or receiving them.

Access networks (LO2)

Access type	Medium	Typical use	Key point
DSL	Telephone copper	Home	Dedicated line to the exchange; speed falls with distance
Cable (HFC)	Coax + fibre	Home	Shared broadcast medium among neighbourhood homes
FTTH	Fibre	Home, business	Highest speeds; passive or active optical networks
Ethernet / Wi-Fi	Twisted pair / radio	Enterprise, campus	LAN access to an edge router
4G/5G cellular	Radio	Mobile	Wide-area wireless access via base stations

Packet switching vs circuit switching (LO3)

- *Packet switching:* messages are broken into packets; resources are used on demand; store-and-forward at each router; queuing and loss are possible. Efficient for bursty data traffic.
- *Circuit switching:* resources (bandwidth, buffers) are reserved end to end for the session using FDM or TDM. Guaranteed rate, but capacity is wasted during silent periods.
- Statistical multiplexing lets packet switching support more users than circuit switching on the same link when users are active only part of the time.

Delay, loss and throughput (LO4)

- Nodal delay = processing + queuing + transmission + propagation.
- *Transmission delay* (L/R) depends on packet length and link rate. *Propagation delay* (d/s) depends on distance and medium speed. Do not confuse them.
- *Traffic intensity* La/R: close to 0 → small queuing delay; approaching 1 → delay grows rapidly; above 1 → queue grows without bound and packets are lost.
- *Throughput* of an end-to-end path is limited by the bottleneck link: min(R1, R2, …, RN).

Layering and encapsulation (LO5)

Layer	PDU	Function	Example protocols
Application	Message	Supports network applications	HTTP, DNS, SMTP
Transport	Segment	Process-to-process delivery	TCP, UDP
Network	Datagram	Host-to-host routing	IP, routing protocols
Link	Frame	Node-to-node transfer on one link	Ethernet, 802.11
Physical	Bits	Moves individual bits on the medium	Copper, fibre, radio

- Each layer adds a header to the data from the layer above (encapsulation); the receiver removes headers in reverse (decapsulation).
- Benefits: modularity, easier maintenance, interoperability between vendors. Drawbacks: duplicated functionality, layers may need information hidden in other layers.

Wireless and mobile networks (LO6–LO10)

- Wireless links differ from wired: decreasing signal strength (path loss), interference, multipath propagation, higher bit error rates, and the hidden terminal problem.

- *802.11 (Wi-Fi):* infrastructure mode uses access points (APs); ad hoc mode has no AP. Uses CSMA/CA with ACKs because collisions cannot be detected while transmitting.
- *CDMA:* each user has a unique code; senders transmit at the same time on the same frequency; the receiver recovers a user's data by taking the inner product with that user's code.
- *4G LTE architecture:* UE (device), eNodeB (base station), MME (mobility and authentication), HSS (subscriber database), S-GW (serving gateway, mobility anchor for data) and P-GW (gateway to the Internet).
- *5G:* higher data rates, lower latency, massive device density; uses millimetre-wave bands with shorter range and a service-based core.
- *Mobility:* handover moves an active connection between base stations within a network; roaming lets a user use a visited network. Indirect routing goes through the home network; direct routing goes straight to the visited network.
- *Mobile IP:* home agent, foreign agent, care-of address (COA). Triangle routing is inefficient; adoption has been limited.

Key terms

Term	Definition
End system	A host (PC, phone, server) at the edge of the network running applications
Store-and-forward	A router must receive the whole packet before transmitting it on the outbound link
Throughput	Rate (bits/s) at which data is actually transferred between end systems
Encapsulation	Adding a layer's header to the payload from the layer above
Handover	Transferring an active mobile connection from one base station to another
Care-of address	Temporary address of a mobile node in a visited network

Worked example: end-to-end delay

A 1,500-byte packet crosses a 10 Mbps link 2,000 km long (propagation speed 2 × 10^8 m/s). Find transmission, propagation and total delay.

1. Convert bytes to bits: 1,500 × 8 = 12,000 bits.

2. Transmission delay = L/R = 12,000 ÷ 10,000,000 = 0.0012 s = **1.2 ms**.
3. Propagation delay = d/s = 2,000,000 ÷ 200,000,000 = 0.01 s = **10 ms**.
4. Total (ignoring processing and queuing) = 1.2 + 10 = **11.2 ms**.
5. Interpretation: on long links, propagation dominates; on slow links, transmission dominates.

Common mistakes

- Forgetting to convert bytes to bits (× 8) or kilo/mega prefixes.
- Describing propagation delay as depending on packet size.
- Listing the seven OSI layers when the question asks for the five-layer Internet stack.
- Confusing handover with roaming.

Revision questions

Level 1 – Recall

1. Define "protocol" and give two examples from different layers. (3)
2. List the four components of nodal delay. (4)
3. Name the PDU at each of the five layers of the Internet stack. (5)
4. State the function of the MME and the P-GW in 4G LTE. (4)

Level 2 – Explain and apply

5. Explain why packet switching supports more users than circuit switching on a bursty link. (4)
6. A file of 8 million bits crosses links of 2 Mbps, 500 kbps and 1 Mbps. Calculate the throughput and approximate transfer time. (4)
7. Explain three ways a wireless link differs from a wired link and the impact of each. (6)
8. Explain the hidden terminal problem and how 802.11 RTS/CTS addresses it. (5)

Level 3 – Analyse, evaluate, scenario

9. A mining company in Limpopo must connect a remote site 60 km from the nearest town to its Johannesburg head office. Evaluate DSL, fixed wireless, 4G LTE and satellite, and recommend one. (12)
10. Analyse how delay and throughput would affect a real-time video consultation between a rural clinic and a city hospital, and recommend two improvements. (10)
11. Evaluate Mobile IP as a mobility solution, including its strengths and practical limitations. (10)

Learning Unit 2: The Application Layer

Kurose & Ross Chapter 2 · Academic weeks 3–4 · LO1–LO7.

Key concepts

Application architectures (LO1, LO7)

- *Client-server:* an always-on server with a fixed IP address serves many clients; clients do not talk directly to each other. Examples: the web, email. Scaling requires data centres.
- *Peer-to-peer (P2P):* peers request and provide service to each other; self-scalable because each new peer adds upload capacity. Examples: BitTorrent. Challenges: security, incentives, peers that join and leave (churn).
- Processes communicate through **sockets**. A process is addressed by IP address + port number (e.g. HTTP server port 80, HTTPS 443, DNS 53).

Transport services for applications (LO2)

Requirement	TCP	UDP
Reliable data transfer	Yes	No
Flow and congestion control	Yes	No
Connection setup	Yes (handshake)	No
Timing / throughput guarantees	No	No
Typical apps	Web, email, file transfer	DNS, streaming, VoIP, gaming

Security (encryption) is added above TCP through TLS.

HTTP (LO3, LO4)

- HTTP is a stateless request-response protocol over TCP. Methods: GET, POST, HEAD, PUT, DELETE. Status codes: 200 OK, 301 Moved Permanently, 400 Bad Request, 404 Not Found, 505 HTTP Version Not Supported.
- *Non-persistent HTTP:* one TCP connection per object, costing 2 RTT + transmission time per object.
- *Persistent HTTP (HTTP/1.1 default):* the connection stays open for multiple objects; with pipelining, requests are sent back to back.
- *HTTP/2:* binary framing, multiplexing of many streams over one connection, request prioritisation and server push — reduces head-of-line (HOL) blocking at the HTTP level. *HTTP/3* runs over QUIC (UDP) to remove TCP-level HOL blocking.

- *Cookies* keep state: a cookie header in the HTTP response, a cookie header in later requests, a cookie file on the user's host, and a back-end database at the site.
- *Web caching (proxy server):* satisfies requests locally, reducing response time and access-link traffic. The *conditional GET* (If-Modified-Since) checks whether the cached copy is still current; a 304 Not Modified reply carries no object.

DNS (LO3)

- Translates hostnames to IP addresses; also provides host aliasing, mail-server aliasing and load distribution. Runs over UDP port 53.
- Hierarchy: root servers → top-level domain (TLD) servers (.com, .za, .ac.za) → authoritative servers. A local DNS server acts as proxy and caches results.
- *Iterative query:* the contacted server replies with the next server to ask. *Recursive query:* the contacted server resolves the name on the requester's behalf.
- Resource records (Name, Value, Type, TTL): A (hostname → IPv4), AAAA (IPv6), NS (domain → authoritative server), CNAME (alias → canonical name), MX (mail server).

Video streaming and CDNs (LO5)

- *DASH:* video is encoded at several bit rates and split into chunks; a manifest file lists them; the client measures bandwidth and picks the best rate chunk by chunk.
- *CDNs* store copies of content close to users. *Enter deep:* many small clusters inside access ISPs (closer, harder to manage, e.g. Akamai). *Bring home:* fewer large clusters at IXPs (easier to manage, higher delay).

Socket programming (LO6)

- *UDP:* no connection; the sender attaches the destination IP and port to each packet.
- *TCP:* the server creates a welcoming socket and listens; when a client connects (three-way handshake), the server creates a new connection socket for that client. Data then flows as a byte stream.

Key terms

Term	Definition
RTT	Round-trip time: time for a small packet to travel from client to server and back
Stateless	The server keeps no information about past client requests
Web cache	Network entity that satisfies HTTP requests on behalf of an origin server

Term	Definition
Authoritative DNS server	Holds the official records for an organisation's hostnames
Manifest file	DASH file listing chunk URLs at each available bit rate
Welcoming socket	TCP server socket that accepts new connection requests

Worked example: P2P vs client-server distribution time

F = 15 Gb, N = 100 peers, server upload us = 30 Mbps, peer download d = 2 Mbps, peer upload u = 700 kbps.

1. Convert F to megabits: 15 Gb = 15,000 Mb.
2. Client-server: D = max(NF/us, F/d) = max(100 × 15,000 ÷ 30, 15,000 ÷ 2) = max(50,000, 7,500) = **50,000 s** (about 13.9 hours).
3. P2P: D = max(F/us, F/d, NF/(us + Nu)) = max(500, 7,500, 1,500,000 ÷ (30 + 70)) = max(500, 7,500, 15,000) = **15,000 s** (about 4.2 hours).
4. Interpretation: client-server time grows linearly with N; P2P grows far more slowly because each peer adds upload capacity.

Common mistakes

- Saying DNS uses TCP for normal queries (it uses UDP; TCP is used for zone transfers and large replies).
- Mixing up iterative and recursive queries.
- Forgetting that HTTP itself is stateless — cookies add state.
- Counting only 1 RTT per object for non-persistent HTTP (it is 2 RTT).

Revision questions

Level 1 – Recall

1. Name three DNS resource record types and what each maps. (3)
2. List four HTTP status codes and their meanings. (4)
3. State the four components of a cookie system. (4)
4. Name the two CDN server-placement philosophies. (2)

Level 2 – Explain and apply

5. Explain how a web cache reduces response time, and the purpose of the conditional GET. (6)
6. A page has a base file and 10 images; RTT = 40 ms. Calculate the response time for non-persistent HTTP and for persistent HTTP with pipelining (ignore transmission time). (6)
7. An institution's 20 Mbps access link carries 18 requests/s for 1 Mb objects. Calculate utilisation, then recalculate with a 50% cache hit rate. (6)
8. Explain step by step how a TCP client and server establish communication using sockets. (6)

Level 3 – Analyse, evaluate, scenario

9. A national broadcaster streams live rugby to 200,000 viewers on mixed fibre and mobile connections. Evaluate how DASH and a CDN would improve viewing quality, and recommend a CDN placement strategy. (15)
10. Trace every application-layer step from typing a URL to the page rendering, including DNS resolution and HTTP exchanges. (12)
11. Analyse the strengths and weaknesses of P2P compared with client-server architecture for distributing a 2 GB software update to 50,000 employees. (12)

Learning Unit 3: The Transport Layer

Kurose & Ross Chapter 3 · Academic weeks 5–6 · LO1–LO7.

Key concepts

Role of the transport layer (LO1)

- Provides *logical communication between processes* on different hosts; the network layer provides logical communication between *hosts*.
- Runs only in end systems, not in routers. Breaks application messages into segments at the sender and reassembles them at the receiver.

Multiplexing and demultiplexing (LO2)

- *Multiplexing:* the sender gathers data from many sockets and adds headers (source and destination ports).
- *Demultiplexing:* the receiver uses header fields to deliver each segment to the correct socket.
- *UDP (connectionless):* socket identified by destination IP + destination port; segments from different sources reach the same socket.
- *TCP (connection-oriented):* socket identified by the four-tuple (source IP, source port, destination IP, destination port); each client gets its own socket on the server.

UDP (LO3)

- Header of 8 bytes: source port, destination port, length, checksum.
- No handshake, no connection state, small header, no congestion control — the application sends as fast as it likes.
- Checksum: sum of 16-bit words with end-around carry, then the 1s complement. Detects (but cannot correct) bit errors.
- Used by DNS, SNMP, streaming, VoIP, gaming and HTTP/3 (QUIC).

Principles of reliable data transfer (LO4)

Mechanism	Purpose
Checksum	Detect bit errors
ACK / NAK	Receiver tells sender whether a packet arrived correctly
Sequence numbers	Detect duplicates and order packets
Timer	Recover from lost packets or ACKs by retransmitting
Window / pipelining	Allow several unacknowledged packets in flight to improve utilisation

- *Stop-and-wait* utilisation = (L/R) ÷ (RTT + L/R), which is very low on high-speed, long-distance links.

Feature	Go-Back-N (GBN)	Selective Repeat (SR)
Sender window	Up to N unACKed packets	Up to N unACKed packets
Receiver window	1 (accepts in order only)	N (buffers out-of-order)
ACKs	Cumulative	Individual per packet
Timer	One, for oldest unACKed packet	One per packet
On loss	Retransmit lost packet and all after it	Retransmit only the lost packet
Best for	Simple receivers, low loss	Lossy links, large windows

TCP (LO5)

- Point-to-point, connection-oriented, full-duplex, reliable, in-order byte stream with flow and congestion control.

- *Segment header* (20 bytes minimum): source/destination port, sequence number (byte number of the first data byte), acknowledgement number (next byte expected), header length, flags (SYN, ACK, FIN, RST, PSH, URG, ECE, CWR), receive window, checksum, urgent pointer, options.
- *Three-way handshake:* client sends SYN (seq = x) → server replies SYN-ACK (seq = y, ack = x + 1) → client sends ACK (ack = y + 1). Closing uses FIN and ACK from each side.
- *Cumulative ACKs*, a single retransmission timer, and *fast retransmit* after three duplicate ACKs.
- *Timeout estimation:* $\text{EstimatedRTT} = (1 - \alpha) \times \text{EstimatedRTT} + \alpha \times \text{SampleRTT}$ ($\alpha = 0.125$); $\text{DevRTT} = (1 - \beta) \times \text{DevRTT} + \beta \times |\text{SampleRTT} - \text{EstimatedRTT}|$ ($\beta = 0.25$); $\text{TimeoutInterval} = \text{EstimatedRTT} + 4 \times \text{DevRTT}$.
- *Flow control:* the receiver advertises its free buffer space (rwnd) so the sender does not overflow it.

Congestion control (LO6, LO7)

- *Causes and costs:* queuing delay grows as arrival rate nears link capacity; retransmissions waste bandwidth; upstream capacity is wasted when packets are dropped downstream.
- *Slow start:* cwnd starts at 1 MSS and doubles every RTT until it reaches ssthresh.
- *Congestion avoidance:* cwnd grows by 1 MSS per RTT (additive increase).
- *On three duplicate ACKs (Reno):* ssthresh = cwnd/2; cwnd = ssthresh + 3 MSS; enter fast recovery (multiplicative decrease).
- *On timeout:* ssthresh = cwnd/2; cwnd = 1 MSS; return to slow start.
- *AIMD* produces the sawtooth pattern. *TCP CUBIC* (Linux default) grows cwnd as a cubic function of time since the last loss, recovering faster on high-bandwidth paths.
- *ECN:* routers mark the IP header instead of dropping packets; the receiver echoes ECE; the sender reduces cwnd and sets CWR.
- *Fairness:* competing TCP flows on a bottleneck converge toward equal shares. UDP flows and parallel TCP connections take more than a fair share.
- *QUIC:* an application-layer protocol over UDP providing reliable, congestion-controlled, encrypted streams with faster connection setup — the basis of HTTP/3.

Key terms

Term	Definition
MSS	Maximum segment size: largest amount of application data in one TCP segment

Term	Definition
cwnd	Congestion window: sender-side limit on unACKed data set by congestion control
rwnd	Receive window: free buffer space advertised by the receiver
ssthresh	Slow-start threshold where TCP switches from exponential to linear growth
Duplicate ACK	An ACK repeating an earlier ACK number, indicating a gap in received data
Pipelining	Sending several packets before receiving their acknowledgements

Worked example: TCP Reno congestion window

cwnd starts at 1 MSS, ssthresh = 16 MSS. A triple duplicate ACK occurs at the end of round 8.

Round	1	2	3	4	5	6	7	8	9	10	11	12
cwnd (MSS)	1	2	4	8	16	17	18	19	9	10	11	12
Phase	SS	SS	SS	SS	SS→CA	CA	CA	CA (loss)	CA	CA	CA	CA

1. Rounds 1–5: slow start, cwnd doubles until it reaches ssthresh = 16.
2. Rounds 6–8: congestion avoidance, +1 MSS per round.
3. End of round 8: triple duplicate ACK; ssthresh = 19 ÷ 2 ≈ 9; cwnd halves to 9 (the textbook figure omits the +3 MSS inflation).
4. Rounds 9–12: linear growth again. A timeout instead would reset cwnd to 1 MSS.

Common mistakes

- Confusing flow control (protects the receiver) with congestion control (protects the network).
- Stating that the TCP ACK number is the last byte received (it is the next byte expected).
- Treating a timeout and a triple duplicate ACK the same way.
- Saying UDP has no error checking (it has a checksum; it has no recovery).

Revision questions

Level 1 – Recall

1. List the four fields of the UDP header. (4)
2. Name the three segments of the TCP handshake and their flags. (3)
3. State the formula for TimeoutInterval. (2)
4. Define cwnd, rwnd and ssthresh. (3)

Level 2 – Explain and apply

5. Calculate the 1s-complement checksum of 0110011001100000, 0101010101010101 and 1000111100001100. (6)
6. RTT = 30 ms, R = 1 Gbps, L = 1,500 bytes. Calculate stop-and-wait utilisation and the window size needed for 90% utilisation. (6)
7. Explain how Go-Back-N and Selective Repeat each recover when packet 3 of a window of 5 is lost. Use timeline diagrams. (8)
8. EstimatedRTT = 80 ms, DevRTT = 6 ms, SampleRTT = 104 ms. Calculate the new TimeoutInterval. (5)

Level 3 – Analyse, evaluate, scenario

9. An e-commerce platform and a video-calling service share one congested 1 Gbps data centre uplink during a Black Friday sale. Analyse the congestion, evaluate how TCP congestion control and ECN respond, and discuss fairness when the video service uses UDP. (15)
10. Evaluate why QUIC was developed and whether it will replace TCP for web traffic. (10)
11. Explain, with a diagram, how TCP delivers reliable in-order data over an unreliable IP layer. (12)

Learning Unit 4: The Network Layer – Data Plane

Kurose & Ross Chapter 4 · Academic weeks 7–8 · LO1–LO7.

Key concepts

Forwarding, routing and the two planes (LO1, LO2, LO3)

- *Forwarding* (data plane): moving a packet from a router's input link to the correct output link. Local, per-router, in nanoseconds, usually in hardware.
- *Routing* (control plane): determining the end-to-end path. Network-wide, in seconds, in software.

- *Traditional approach:* routing algorithms run in every router and interact to build each forwarding table.
- *SDN approach:* a logically centralised remote controller computes and installs forwarding tables in the routers. Benefits: easier management, programmability, fewer misconfigurations, faster innovation.
- *Destination-based forwarding* uses **longest prefix matching**: when several table entries match, the most specific (longest) prefix wins.

Network service models (LO4)

- Possible services: guaranteed delivery, guaranteed delivery with bounded delay, in-order delivery, guaranteed minimum bandwidth, security.
- The Internet offers **best-effort** service only: no guarantees of delivery, order, delay or bandwidth. Its simplicity, combined with over-provisioning and end-to-end mechanisms (TCP), made it successful.

Inside a router (LO5)

Component	Role
Input ports	Terminate the link, perform link-layer processing, look up the forwarding table, queue if the fabric is busy
Switching fabric	Transfers packets from input to output ports: via memory, a shared bus, or an interconnection network (crossbar)
Output ports	Buffer packets, schedule them and transmit onto the link
Routing processor	Runs control-plane functions and computes or receives the forwarding table

- *Input queuing* occurs when the fabric is slower than the combined input rate; *head-of-line (HOL) blocking* means a packet at the front of a queue blocks those behind it, even if their outputs are free.
- *Output queuing* occurs when packets arrive faster than the output link rate; buffer overflow causes drops (drop-tail or active queue management such as RED).

Scheduling	How it works	Strength	Weakness
FIFO	Transmit in order of arrival	Simple	No differentiation

Scheduling	How it works	Strength	Weakness
Priority	Highest-priority class served first	Low delay for critical traffic	Low classes can starve
Round robin	Classes served in turn	Fair between classes	Ignores class importance
Weighted fair queuing (WFQ)	Each class gets a weighted share	Guaranteed minimum bandwidth	More complex

IPv4, addressing and DHCP (LO6)

- IPv4 header (20 bytes without options): version, header length, type of service, datagram length, identifier, flags, fragment offset, TTL, upper-layer protocol, header checksum, source IP, destination IP.
- *Fragmentation:* routers split datagrams larger than the link MTU; fragments share an identifier, carry an offset in 8-byte units, and the MF flag is 1 on all but the last. The destination host reassembles them.
- *CIDR notation* a.b.c.d/x: the first x bits are the network (subnet) part.
- Usable hosts in a subnet = 2^(32 − x) − 2 (network and broadcast addresses excluded).
- *DHCP* (plug-and-play addressing): **D**iscover → **O**ffer → **R**equest → **A**CK (DORA), over UDP ports 67/68. It also supplies the subnet mask, default gateway and DNS server.
- *NAT:* a router translates many private addresses (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) to one public address using port numbers, kept in a NAT translation table. Pros: saves IPv4 addresses, hides internal hosts. Cons: breaks end-to-end principle, complicates P2P and incoming connections, routers handle layer-4 information.

IPv6 (LO6)

- 128-bit addresses; fixed 40-byte header; flow label; no fragmentation at routers (ICMPv6 "Packet Too Big" instead); no header checksum; options moved to extension headers.
- *Transition:* **dual stack** (devices run both IPv4 and IPv6) and **tunnelling** (IPv6 datagrams carried as the payload of IPv4 datagrams across IPv4-only routers).

Generalised forwarding and middleboxes (LO7)

- *Match-plus-action:* a flow table entry matches header fields across layers (MAC, IP, port, VLAN) and applies an action (forward, drop, modify, send to controller), with counters for statistics. OpenFlow is the standard protocol.
- One flow table can act as a router (match destination IP), switch (match MAC), firewall (match ports and drop) or NAT (match and rewrite addresses).

- *Middleboxes* perform functions other than standard IP forwarding: NAT, firewalls, intrusion detection systems (IDS), load balancers, caches.

Key terms

Term	Definition
Longest prefix match	Forwarding rule that picks the most specific matching table entry
MTU	Maximum transmission unit: largest link-layer frame payload a link can carry
TTL	Field decremented at each router; datagram discarded when it reaches 0
Subnet	Group of interfaces that can reach each other without passing through a router
Flow table	SDN table of match-plus-action rules installed by the controller
Middlebox	Device performing functions beyond IP forwarding on the data path

Worked example: VLSM subnetting

Divide 192.168.10.0/24 for four subnets of 50, 25, 12 and 6 hosts.

1. Sort largest first and find the smallest block that fits each (hosts + 2).
2. 50 hosts → 64 addresses (/26); 25 → 32 (/27); 12 → 16 (/28); 6 → 8 (/29).
3. Allocate in order:

Hosts needed	Network	Prefix	Usable range	Broadcast
50	192.168.10.0	/26	.1 – .62	.63
25	192.168.10.64	/27	.65 – .94	.95
12	192.168.10.96	/28	.97 – .110	.111
6	192.168.10.112	/29	.113 – .118	.119

4. Remaining space for growth: 192.168.10.120 – .255.

Common mistakes

- Forgetting to subtract 2 when counting usable hosts.
- Allocating small subnets first, which leaves gaps and misaligned boundaries.
- Expressing fragment offset in bytes instead of 8-byte units.
- Saying IPv6 routers fragment datagrams.

Revision questions

Level 1 – Recall

1. Distinguish forwarding from routing in one sentence each. (2)
2. Name the four DHCP messages in order. (4)
3. List three differences between IPv4 and IPv6 headers. (3)
4. Name four types of middlebox. (4)

Level 2 – Explain and apply

5. For 172.16.45.200/20, find the network address, broadcast address, mask and usable hosts. (5)
6. A 4,000-byte datagram (20-byte header) crosses a link with an MTU of 1,500 bytes. Give the length, offset and MF flag of each fragment. (6)
7. Explain how NAT translates outgoing and returning traffic, using a translation table. (6)
8. Compare FIFO, priority and weighted fair queuing for a network carrying VoIP and file downloads. (6)

Level 3 – Analyse, evaluate, scenario

9. A hospital using 10.30.0.0/16 needs subnets for Wards (400 hosts), Admin (150), Radiology (60), Pharmacy (20) and Management (10). Design a VLSM plan and explain how DHCP and NAT fit into the design. (15)
10. Compare traditional destination-based forwarding with SDN generalised forwarding, and evaluate whether a mid-sized ISP should adopt SDN. (12)
11. Evaluate dual stack and tunnelling as IPv6 transition strategies for a South African university. (10)

Learning Unit 5: The Network Layer – Control Plane

Kurose & Ross Chapter 5 · Academic weeks 8–10 · LO1–LO7.

Key concepts

Role of the control plane (LO1)

- Computes the forwarding tables used by the data plane. Two approaches: *per-router control* (each router runs the routing algorithm) and *logically centralised control* (an SDN controller computes and distributes tables).
- Routing algorithms are classified as *centralised (global)* vs *decentralised*, *static* vs *dynamic*, and *load-sensitive* vs *load-insensitive*.

Link-state vs distance-vector routing (LO3)

- *Link-state (Dijkstra):* every router learns the full topology and link costs by flooding link-state advertisements, then computes least-cost paths to all destinations. Complexity $O(n^2)$ in the basic form.
- *Distance-vector (Bellman-Ford):* each router knows only its neighbours' distance vectors. $Dx(y) = \min$ over neighbours v of $\{c(x,v) + Dv(y)\}$. Iterative, asynchronous and distributed.
- *Count-to-infinity:* when a link cost rises, DV routers may loop and increase costs slowly ("bad news travels slowly"). *Poisoned reverse* (advertising infinity back to the neighbour you route through) fixes two-node loops but not larger ones.

Criterion	Link-state	Distance-vector
Knowledge	Whole topology	Neighbours only
Message complexity	High (flooding to all routers)	Low (neighbours only)
Convergence	Fast, $O(n^2)$	Variable; count-to-infinity possible
Robustness	Errors stay local to each router's own computation	Errors propagate through the network
Example	OSPF, IS-IS	RIP

Intra-AS routing: OSPF (LO2)

- Link-state protocol; uses Dijkstra; floods LSAs to all routers in the AS directly over IP.
- Features: authentication of messages, multiple equal-cost paths, integrated unicast and multicast support, and **hierarchical areas** with a backbone area connecting all others through area border routers.
- Administrators set link costs (e.g. inversely proportional to bandwidth).

Inter-AS routing: BGP (LO2, LO4)

- The Internet is a network of Autonomous Systems (ASs); BGP glues them together. It is a **path-vector** protocol over TCP port 179.

- *eBGP:* sessions between routers in different ASs. *iBGP:* sessions between routers inside the same AS, to spread externally learned routes.
- *Route = prefix + attributes.* AS-PATH lists the ASs traversed (also used to detect loops). NEXT-HOP gives the IP of the router interface that begins the AS-PATH.
- *Route selection, in order:* (1) highest LOCAL-PREF (policy), (2) shortest AS-PATH, (3) closest NEXT-HOP (**hot-potato routing**), (4) further tie-breakers such as BGP identifiers.
- *Policy:* a customer network that is multi-homed to two providers does not advertise routes learned from one provider to the other, so it never carries transit traffic between them.
- *Why separate intra- and inter-AS routing?* Scale (no router can hold all routes), administrative autonomy, policy control between organisations, and performance optimisation inside each AS.

IP anycast (LO5)

- *Unicast:* one sender to one specific receiver. *Multicast:* one sender to a group of receivers. *Anycast:* the same IP address is advertised from several locations, and BGP delivers each packet to the nearest one.
- Used by DNS root servers and CDNs to improve latency and resilience.

Network management (LO6)

- Components: *managing server* (network operations centre), *managed devices* (routers, switches, hosts), *MIB* data (status and configuration objects), *agents* on each device, and a *network management protocol*.
- *SNMP* (over UDP): GetRequest, GetNextRequest, GetBulkRequest, SetRequest, Response and Trap (an unsolicited alert from agent to manager). Best for monitoring; weak for configuring many devices at once.
- *NETCONF* (over a secure, TCP-based session): manages whole device configurations with atomic commit and rollback, encoded in XML. *YANG* is the data-modelling language that defines the structure of that configuration data.

ICMP (LO7)

- Used by hosts and routers to report errors and exchange diagnostic information. Messages are carried inside IP datagrams and contain a type and code, plus the header of the datagram that caused the error.
- *Ping:* sends ICMP echo request (type 8) and receives echo reply (type 0).
- *Traceroute:* sends datagrams with TTL = 1, 2, 3 …; each router that discards one returns ICMP "TTL expired" (type 11), revealing its address and RTT; the destination finally returns "port unreachable" (type 3).

Key terms

Term	Definition
Autonomous System (AS)	Group of routers under one administrative control, identified by an AS number
LSA	Link-state advertisement flooded by a router describing its links and costs
Hot-potato routing	Sending traffic to the gateway with the lowest intra-AS cost, regardless of the rest of the path
LOCAL-PREF	BGP attribute set by local policy to prefer one route over others
MIB	Management information base: the collection of objects describing a device's state
Anycast	Same IP address served from multiple locations; packets reach the nearest

Worked example: Dijkstra's algorithm

Links: u–v 2, u–w 5, u–x 1, v–w 3, v–x 2, x–w 3, x–y 1, w–y 1, w–z 5, y–z 2. Find least-cost paths from u.

Step	N′ (settled)	D(v), p(v)	D(w), p(w)	D(x), p(x)	D(y), p(y)	D(z), p(z)
0	u	2, u	5, u	1, u	∞	∞
1	ux	2, u	4, x	–	2, x	∞
2	uxy	2, u	3, y	–	–	4, y
3	uxyv	–	3, y	–	–	4, y
4	uxyvw	–	–	–	–	4, y
5	uxyvwz	–	–	–	–	–

- Results: x = 1 (u→x), v = 2 (u→v), y = 2 (u→x→y), w = 3 (u→x→y→w), z = 4 (u→x→y→z).
- Forwarding table at u: destination v → link (u, v); destinations x, y, w, z → link (u, x).
- Ties: at step 1, v and y both cost 2; either may be chosen first without changing the final result.

Common mistakes

- Calling BGP a distance-vector protocol (it is path-vector).

- Forgetting to update a node's cost when a cheaper path appears through a newly settled node.
- Confusing iBGP (inside one AS) with an intra-AS routing protocol such as OSPF.
- Stating that ICMP runs over TCP or UDP.

Revision questions

Level 1 – Recall

1. State the Bellman-Ford equation. (2)
2. List the BGP route-selection rules in order. (4)
3. Name the components of a network management framework. (5)
4. State the ICMP types used by ping and traceroute. (3)

Level 2 – Explain and apply

5. Three nodes have costs x–y = 2, y–z = 1, x–z = 7. Show the distance-vector tables at each node until convergence. (8)
6. Explain count-to-infinity and how poisoned reverse reduces it. (6)
7. Explain the difference between eBGP and iBGP, and how a route learned at a gateway reaches internal routers. (6)
8. Compare SNMP with NETCONF/YANG for configuring 500 routers. (6)

Level 3 – Analyse, evaluate, scenario

9. A Johannesburg ISP (AS 65010) has two upstream providers and peers with a content provider at an IXP. Explain how it would use LOCAL-PREF, AS-PATH and policy to send streaming traffic via the IXP and avoid carrying transit traffic. (15)
10. Evaluate link-state and distance-vector algorithms for a national retail chain with 300 branch routers, and recommend one. (12)
11. Analyse how IP anycast improves the resilience of DNS root servers during a distributed denial-of-service attack. (8)

Learning Unit 6: The Link Layer and LANs

Kurose & Ross Chapter 6 · Academic weeks 11–12 · LO1–LO7.

Key concepts

Link-layer services and implementation (LO1, LO2)

- Moves a datagram from one node to an *adjacent* node over a single link. Different links may use different protocols along one path (Ethernet, then 802.11).

- Services: *framing*, *link access* (MAC protocol for shared media), *reliable delivery* (used on error-prone wireless links), *error detection and correction*, *flow control* between adjacent nodes, and *half- or full-duplex* operation.
- Implemented mainly in the **network interface card (NIC)** — hardware plus firmware — with some functions in host software.
- *Interaction with other layers:* the network layer hands down a datagram; the link layer encapsulates it in a frame addressed with MAC addresses; the physical layer sends the bits. At each router the frame is removed and a new frame is built for the next link, while the IP datagram travels end to end.
- *Broadcast links* (shared medium: classic Ethernet, Wi-Fi, cable) need multiple access protocols. *Point-to-point links* (one sender, one receiver: switch-to-host Ethernet, PPP) do not.

Error detection and correction (LO3)

Technique	How it works	Detects	Where used
Single-bit parity	One bit makes the count of 1s even (or odd)	Any odd number of bit errors	Simple links, memory
Two-dimensional parity	Parity per row and column	Detects and corrects single-bit errors	Teaching example, some storage
Internet checksum	1s-complement sum of 16-bit words	Many errors; weaker	Transport and network layers (in software)
CRC	Remainder of modulo-2 division by generator G (r + 1 bits)	All burst errors ⩽ r bits, plus most longer ones	Link layer (Ethernet, Wi-Fi), in hardware

Multiple access protocols (LO4)

Class	How it works	Examples	Strength	Weakness
Channel partitioning	Divide the channel into fixed slots or frequencies	TDMA, FDMA, CDMA	No collisions; fair under heavy load	Wasteful when few nodes are active

Class	How it works	Examples	Strength	Weakness
Random access	Transmit at full rate; recover from collisions	ALOHA, slotted ALOHA, CSMA, CSMA/CD, CSMA/CA	Efficient under light load; decentralised	Collisions waste capacity under heavy load
Taking turns	Nodes take turns through polling or a token	Polling, token passing, Bluetooth	Efficient under both loads	Polling and token overhead; single point of failure

- *Slotted ALOHA* maximum efficiency = 1/e ≈ 37%; *pure ALOHA* = 1/(2e) ≈ 18%.
- *CSMA/CD (classic Ethernet):* listen before transmitting; abort when a collision is detected; send a jam signal; then use **binary exponential backoff** — after the nth collision choose K at random from {0, 1, …, 2^n − 1} and wait K × 512 bit times.
- *CSMA/CA (802.11):* collisions cannot be detected while transmitting, so stations wait random backoff times, use link-layer ACKs and optional RTS/CTS reservation.

Addressing, ARP and Ethernet (LO5)

- *MAC address:* 48-bit, burned into the NIC, flat (not hierarchical), written as 1A-2F-BB-76-09-AD. Broadcast address: FF-FF-FF-FF-FF-FF.
- *ARP* resolves an IP address on the same subnet to a MAC address: broadcast ARP query → unicast ARP reply → result cached in the ARP table with a TTL. To reach a host on another subnet, the sender ARPs for its **default gateway's** MAC address, not the destination's.
- *Ethernet frame:* preamble (8 bytes), destination MAC (6), source MAC (6), type (2, e.g. IPv4 or ARP), data (46–1,500 bytes), CRC (4). Connectionless and unreliable: bad frames are dropped, not ACKed.
- *Switches:* store-and-forward layer-2 devices; transparent to hosts; plug-and-play; *self-learning* — they record the source MAC and arrival port of each frame in the switch table. Known destination on another port → forward; same port → filter (drop); unknown → **flood** to all other ports.

Feature	Switch	Router
Layer	Link (2)	Network (3)
Addresses	MAC	IP
Configuration	Plug-and-play, self-learning	Requires IP configuration and routing

Feature	Switch	Router
Topology	Spanning tree only; broadcast storms possible	Any topology; firewall protection against storms
Table	Switch table (learned)	Forwarding table (computed by routing)

VLANs and data centres (LO6)

- *Problems with flat LANs:* all broadcast traffic (ARP, DHCP) crosses the whole LAN; no isolation between departments (security and privacy); inefficient use of switches; moving users means rewiring.
- *Port-based VLANs:* switch ports are grouped so one physical switch acts as several virtual switches. Traffic between VLANs must pass through a router (or layer-3 switch).
- *Trunk ports* carry frames for multiple VLANs between switches; **802.1Q** adds a 4-byte tag containing the 12-bit VLAN ID (up to 4,094 VLANs).
- *Data centre networks:* hierarchical topology of top-of-rack (TOR) switches, tier-2 and tier-1 switches, border routers and load balancers. Rich interconnection (e.g. Clos or fat-tree designs) gives multiple equal-cost paths, high throughput and redundancy. Trends: SDN control, virtualisation and container-based application hosting.

Supporting services: a day in the life of a web request (LO7)

1. *DHCP* (over UDP, IP broadcast, Ethernet broadcast): the laptop obtains its IP address, subnet mask, default gateway and DNS server.
2. *ARP:* the laptop learns the gateway router's MAC address.
3. *DNS* (over UDP): the laptop resolves the website's hostname to an IP address, sending the query through the gateway.
4. *Routing:* intra-domain (OSPF) and inter-domain (BGP) routes carry the datagram across ISPs to the server.
5. *TCP:* three-way handshake with the web server.
6. *HTTP:* GET request sent; response returned and rendered by the browser.

Key terms

Term	Definition
Frame	Link-layer PDU encapsulating a network-layer datagram

Term	Definition
NIC	Network interface card implementing link and physical layers
Collision	Two or more simultaneous transmissions on a shared channel
Flooding	Sending a frame out of all switch ports except the arrival port
VLAN	Logical LAN defined in switch software, independent of physical location
Trunk port	Switch port carrying tagged frames for several VLANs

Worked example: CRC calculation

Data D = 101110, generator G = 1001 (r = 3).

1. Append r = 3 zeros to D: 101110**000** (9 bits).
2. Divide by 1001 using modulo-2 (XOR) arithmetic, bringing down one bit at a time:
   a. First 4 bits 1011 XOR 1001 = 0010.
   b. Bring down 1 → 0101: leading bit 0, so no XOR.
   c. Bring down 0 → 1010; XOR 1001 = 0011.
   d. Bring down 0 → 0110: leading bit 0, so no XOR.
   e. Bring down 0 → 1100; XOR 1001 = 0101.
   f. Bring down 0 → 1010; XOR 1001 = 0011. All bits used, so remainder R = **011**.
3. Transmit D followed by R: **101110011**.
4. Receiver check: dividing 101110011 by 1001 gives remainder 000, so no error is detected. Any non-zero remainder indicates an error.

Common mistakes

- Saying ARP finds the MAC address of a remote host on another subnet (it finds the gateway's).
- Confusing CSMA/CD (wired Ethernet) with CSMA/CA (Wi-Fi).
- Forgetting that switches flood unknown destinations.
- Using ordinary subtraction instead of XOR in CRC division.

Revision questions

Level 1 – Recall

1. List four services the link layer can provide. (4)
2. Name the three classes of multiple access protocols with one example each. (6)
3. List the fields of an Ethernet frame. (6)
4. State the purpose of the 802.1Q tag. (2)

Level 2 – Explain and apply

5. Using even two-dimensional parity, build the parity matrix for 1110 0110 1001 0101 and show how a single-bit error is corrected. (6)
6. Calculate the CRC for D = 110101 and G = 1011, and state the transmitted bits. (6)
7. Host A (subnet 1) sends to Host B (subnet 2) through router R. Describe each ARP exchange and the MAC and IP addresses in each frame. (8)
8. Explain how a switch builds its table and handles a frame for an unknown destination. (5)

Level 3 – Analyse, evaluate, scenario

9. A Cape Town law firm has 250 staff in Litigation, Finance, HR and a Guest Wi-Fi network on one flat LAN. Auditors report guests can see Finance traffic. Analyse the problems and design a VLAN solution with trunking and inter-VLAN routing. (15)
10. Evaluate which multiple access class best suits (a) a busy factory sensor network and (b) a lightly used home network. (10)
11. Trace every protocol involved from a student plugging a laptop into a campus Ethernet port to a web page displaying. (15)

Formula sheet and quick reference

Learn these from memory: the exam is closed book and no formula sheet is supplied.

Formulas

Topic	Formula	Units and notes
Transmission delay	d_trans = L / R	L in bits, R in bits/s
Propagation delay	d_prop = d / s	d in metres, s ≈ 2 × 10^8 m/s
Nodal delay	d_nodal = d_proc + d_queue + d_trans + d_prop	Sum of four components

Topic	Formula	Units and notes
End-to-end delay (N links, no queuing)	N × (d_proc + d_trans + d_prop)	Store-and-forward at each hop
Traffic intensity	La / R	a = packets/s; keep below 1
Throughput	min(R1, R2, …, RN)	Bottleneck link
Non-persistent HTTP	2 RTT + transmission time per object	Plus parallelism if allowed
Client-server distribution	D = max(NF / us, F / dmin)	Grows linearly with N
P2P distribution	D = max(F / us, F / dmin, NF / (us + Σui))	Grows slowly with N
Stop-and-wait utilisation	U = (L/R) / (RTT + L/R)	Pipelined: multiply numerator by window N
Estimated RTT	(1 − α) × EstimatedRTT + α × SampleRTT	α = 0.125
RTT deviation	(1 − β) × DevRTT + β × |SampleRTT − EstimatedRTT|	β = 0.25
TCP timeout	EstimatedRTT + 4 × DevRTT	–
Usable hosts per subnet	2^(32 − prefix) − 2	Excludes network and broadcast
Fragment offset	Byte position of data ÷ 8	Data per fragment = MTU − 20, multiple of 8
Bellman-Ford	Dx(y) = min_v { c(x,v) + Dv(y) }	Over all neighbours v
Slotted ALOHA efficiency	Np(1 − p)^(N − 1) → 1/e ≈ 0.37	Pure ALOHA → 1/(2e) ≈ 0.18
CSMA/CD backoff	K from {0, …, 2^n − 1}; wait K × 512 bit times	n = number of collisions (max 10)

Protocols at a glance

Protocol	Layer	Transport / port	Purpose
HTTP / HTTPS	Application	TCP 80 / 443	Web transfer

Protocol	Layer	Transport / port	Purpose
HTTP/3	Application	QUIC over UDP 443	Web transfer without TCP HOL blocking
DNS	Application	UDP 53 (TCP for large replies)	Name resolution
DHCP	Application	UDP 67/68	Automatic IP configuration
SNMP	Application	UDP 161/162	Monitoring and traps
NETCONF	Application	TCP (SSH) 830	Configuration management
BGP	Application (routing)	TCP 179	Inter-AS routing
TCP	Transport	–	Reliable byte stream
UDP	Transport	–	Connectionless datagrams
IP (v4/v6)	Network	–	Addressing and forwarding
ICMP	Network	Inside IP	Errors and diagnostics
OSPF	Network (routing)	Directly over IP	Intra-AS link-state routing
ARP	Link / network boundary	Ethernet broadcast	IP → MAC resolution
Ethernet	Link	–	Wired LAN frames
802.11	Link	–	Wireless LAN

Common ports and addresses

- Private IPv4 ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- Loopback: 127.0.0.1 (IPv4), ::1 (IPv6). Limited broadcast: 255.255.255.255. Ethernet broadcast: FF-FF-FF-FF-FF-FF.
- Prefix to hosts: /24 = 254 · /25 = 126 · /26 = 62 · /27 = 30 · /28 = 14 · /29 = 6 · /30 = 2.

Self-assessment and exam-day checklist

Tick each statement only when you can do it from memory, without notes. Anything unticked is your revision priority.

Self-assessment by learning unit

LU1 – Internet, wireless and mobile

- [ ] I can describe the Internet's ISP hierarchy, IXPs and content-provider networks.
- [ ] I can compare packet and circuit switching and explain statistical multiplexing.
- [ ] I can calculate transmission, propagation and end-to-end delay, and throughput.
- [ ] I can name the five layers, their PDUs and explain encapsulation.
- [ ] I can explain CDMA, the 4G LTE components and how handover works.

LU2 – Application layer

- [ ] I can compare client-server and P2P, and calculate distribution times.
- [ ] I can calculate HTTP response times for non-persistent and persistent connections.
- [ ] I can explain cookies, web caching, the conditional GET and HTTP/2.
- [ ] I can trace a DNS lookup through root, TLD and authoritative servers.
- [ ] I can explain DASH, CDNs and TCP socket programming.

LU3 – Transport layer

- [ ] I can explain multiplexing, demultiplexing and the UDP header and checksum.
- [ ] I can compare Go-Back-N and Selective Repeat with timeline diagrams.
- [ ] I can describe the TCP handshake, header fields, ACKs and timeout calculation.
- [ ] I can trace cwnd through slow start, congestion avoidance and fast recovery.
- [ ] I can explain ECN, fairness and QUIC.

LU4 – Network layer data plane

- [ ] I can distinguish forwarding from routing and traditional control from SDN.
- [ ] I can describe router components, HOL blocking and scheduling disciplines.
- [ ] I can design a VLSM plan and work out network and broadcast addresses.
- [ ] I can calculate IP fragments and explain DHCP and NAT.
- [ ] I can compare IPv4 and IPv6 and explain match-plus-action forwarding.

LU5 – Network layer control plane

- [ ] I can run Dijkstra's algorithm and build a forwarding table.
- [ ] I can run distance-vector updates and explain count-to-infinity.
- [ ] I can explain OSPF areas, eBGP/iBGP and the BGP route-selection rules.
- [ ] I can explain anycast, SNMP, NETCONF/YANG and ICMP (ping and traceroute).

LU6 – Link layer and LANs

- [ ] I can calculate parity and CRC bits and explain what each detects.

- [ ] I can compare the three classes of multiple access protocols and explain CSMA/CD.
- [ ] I can explain ARP within and across subnets and the Ethernet frame format.
- [ ] I can explain switch self-learning and compare switches with routers.
- [ ] I can design a VLAN solution and trace a full web request from DHCP to HTTP.

Exam-day checklist

1. **Before you start:** read the whole paper in the first 5 minutes and mark the questions you can answer best.
2. **Read the task word:** define, describe, explain, compare, analyse, evaluate and recommend each need a different depth.
3. **Match marks to points:** one well-developed point per mark; spend about 1.2 minutes per mark.
4. **Show working in calculations:** formula, substitution with units, answer and a one-line interpretation. Method marks are awarded even when the final answer is wrong.
5. **Use the scenario:** in long questions, refer to the organisation, its needs and its numbers in every paragraph. Generic answers lose application marks.
6. **Use tables and diagrams:** comparisons in a table; handshakes, timelines and topologies as labelled diagrams.
7. **End evaluate and recommend questions with a justified judgement** that names trade-offs.
8. **Leave 10 minutes** to check numbering, units and unanswered parts.